

3D Printing Strategies for Bioengineering Human Cornea

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Over 4.2 million people worldwide suffer from vision impairment related to corneal diseases, with the shortage of donor corneas limiting surgical interventions. The development of 3D printing provides a promising solution for corneal tissue engineering, offering precise control over dimensionality, structural organization, and cell–matrix interactions. The choice of biomaterials and printing strategies critically determines the properties of 3D-printed corneal constructs. While prior reviews have summarized general bioinks and techniques, few have addressed cornea-specific benchmarks such as transparency, curvature, and mechanical robustness. This review provides a comprehensive discussion on the current study of 3D printed cornea, with a focus on replicating the biomechanical and optical properties of native human cornea. Key limitations in current fabrication methods are outlined, and recent advancements in 3D bioprinting technologies—many of which have been successfully applied in other tissue engineering contexts—are reviewed for their potential in producing anisotropic, multilayered corneal constructs with high resolution and fidelity. 3D printing human cornea is believed to have strong potential in generating scalable and clinically relevant solutions to overcome the global donor cornea tissue shortage, and 3D printing holds a central role in their development.

incoming light, it provides nearly two-thirds of the eye's total focusing power.^[1] Biochemically, it consists largely of water (around 78%) and collagen (≈15%), along with specialized cells and regulatory proteins that maintain its clarity and structure.^[2] When the cornea becomes diseased or damaged, its transparency and biomechanical function can be compromised, resulting in visual impairment. For example, stromal injury may lead to permanent scarring, endothelial cell loss can cause fluid accumulation and swelling, and infections or chronic inflammation can further deteriorate corneal integrity. Globally, more than 4.2 million people are affected by corneal blindness or severe visual impairment, with corneal transplantation remaining the gold-standard intervention to restore sight.^[3]

However, a global donor cornea shortage has limited the use of corneal transplantation to patients. Artificial corneal alternatives are developed to address the issue. Keratoprotheses, derived from the Greek words for “cornea” and “addition,”

represent the most commonly used artificial corneal replacement in severe corneal disease cases.^[4] These prosthetic devices are typically composed of medical-grade plastic, titanium, or other polymers, with or without donor tissue integration. **Figure 1** shows the schematic illustrations of 4 commonly used keratoprosthesis: Osteo-odonto-keratoprosthesis (OOKP), Boston

1. Introduction

The cornea forms the transparent, dome-shaped front surface of the eye, acting both as a physical shield against external insults such as pathogens, trauma, and UV radiation, and as a key optical component for vision. By transmitting and refracting

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DOI: 10.1002/adhm.202502767

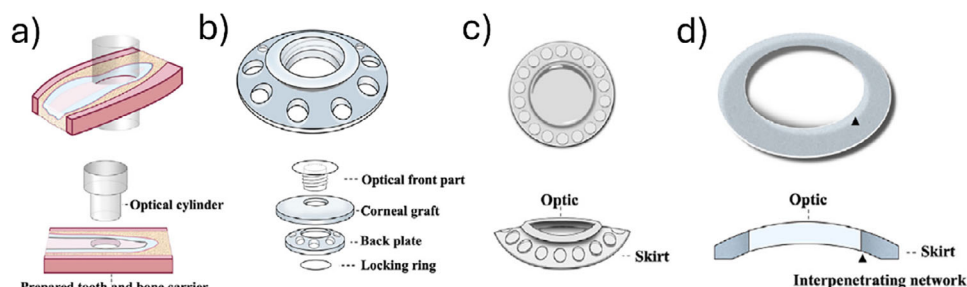


Figure 1. Schematic illustrations of 4 commonly used keratoprosthesis: a) OOKP, b) Boston KPro, c) Keraclear, and d) AlphaCo. Reproduced with permission^[4] Copyright 2024, AccScience Publishing.

Keratoprosthesis (KPro), Keraclear, and AlphaCo.^[4] However, materials used in keratoprosthesis could exhibit poor biocompatibility, higher rejection rate and limited long-term durability, making them suboptimal replacements for natural corneal tissue. In addition, these keratoprosthesis are made using a classic manufacturing approach involving moulding, and are not customized to an individual's needs.

Bioengineered human cornea comprising biomaterials and appropriate distribution of cells has emerged in order to address the limitations of available keratoprosthesis. 3D printing offers a promising approach for fabricating customized 3D tissue with tailored properties, including mechanical properties, biocompatibility, flexibility, reproducibility, and scalability.^[5,6] A PubMed search on 3D printing cornea revealed a total of 40 publications with 33 (82.5%) being research articles, and 7 (17.5%) review articles (**Figure 2a**). The first 3D print cornea study was published in April 2018 and used laser-assisted printing of human embryonic stem cell-derived limbal epithelial stem cells and human adipose tissue-derived stem cells to fabricate epithelium-mimicking structures and stroma-mimicking structures, respectively.^[7] The number of published articles increased significantly in 2024, which shows great interest from researchers in accelerating 3D printing cornea regeneration (**Figure 2b**).

Both the selection of biomaterials and the choice of printing methods are central in shaping the physicochemical properties of

the printed constructs. Previous reviews have discussed printing strategies, bioink types, and their influence on cell behaviour, but critical gaps remain in addressing corneal-specific requirements, especially in replicating the morphology and strength of biofabricated cornea.^[2,3,26] In this review, we will examine the unique biomechanical properties of the cornea, including those uncommonly discussed (e.g. refractive index), and the pros and cons of each published 3D printing cornea approach. In addition, we will review the recent advancements in 3D bioprinting technologies for their potential in producing anisotropic, multilayered corneal constructs with high resolution and fidelity. We aim to provide insights into a key question: Is the fabrication of a bioengineered cornea using 3D printing feasible?.

2. Corneal Structure and Its Unique Biomechanical Properties

The cornea is a dome-shaped tissue, positioned at the front of the eye. The cornea is highly transparent, allowing 90% of visible light to travel through, acting as the primary refractive surface, bending light to focus it on the retina.^[1] The refractive power of the human cornea varies across its depth, from 1.400 at the epithelium, 1.380 at Bowman's layer, 1.369 in the mid stroma, and 1.373 at the endothelium. The overall average refractive power of the human cornea is 1.37.^[8] In addition to being the main

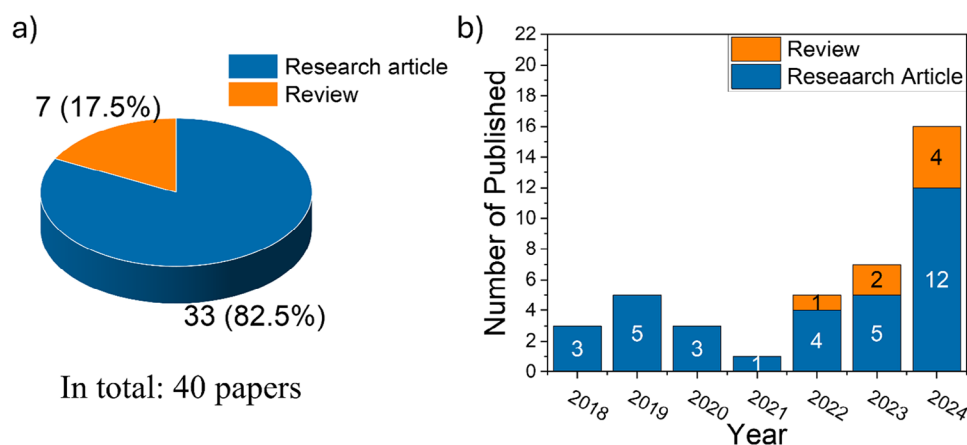


Figure 2. Publication analysis of 3D-printed cornea research. a) Distribution of publications by type, showing the proportion of research articles and review papers (total: 40 papers). b) Annual publication trends from 2018 to 2024, indicating the number of research articles and review papers published each year.

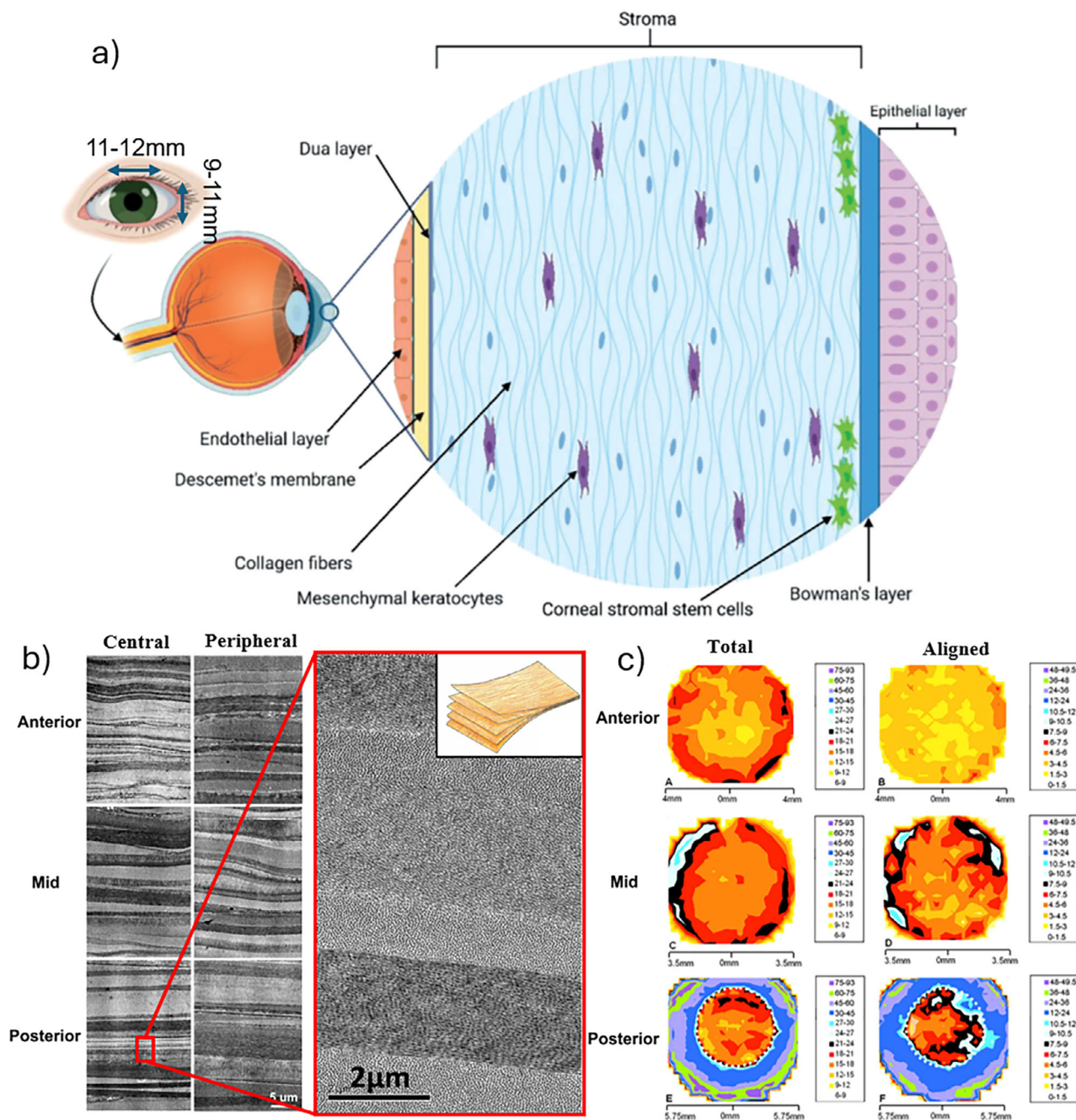


Figure 3. Structural and ultrastructural features of the human cornea. a) Schematic representation of corneal anatomy, including epithelial, stromal, and endothelial layers, with key structural components highlighted. Reproduced with permission^[4] Copyright 2024, AccScience Publishing. b) Transmission electron microscope pictures show collagen fibril arrangements in the anterior, mid, and posterior stroma at both central and peripheral regions. The orthogonal lamellar pattern resembles a plywood-like structure, where adjacent layers run in alternating orientations. Reproduced with permission^[18,25] Copyright 2024, Elsevier Ltd. and Copyright 2024, Springer Nature. c) Contour maps of x-ray scattering intensity illustrating the distribution of total fibrillar collagen (left) and the degree of fibril alignment (right) across anterior, mid, and posterior stromal layers. Higher values correspond to increased collagen content and alignment. Reproduced with permission^[19] Copyright 2009, Association for Research in Vision and Ophthalmology.

light refractive tissue in the eye, the cornea serves as the guardian to protect the eye from external substances, mechanical damage, and infections.

Structurally, it is composed of five distinct layers: epithelium, Bowman's layer, stroma, Descemet's membrane, and endothe-

lium (Figure 3a).^[1,9] The Young's modulus of native human cornea is around 115–520 kPa, with the value varying based on the age, environmental temperature and humidity when testing.^[10–12] The horizontal diameter of the adult cornea ranges from 11 to 12 mm, while the vertical diameter varies between 9

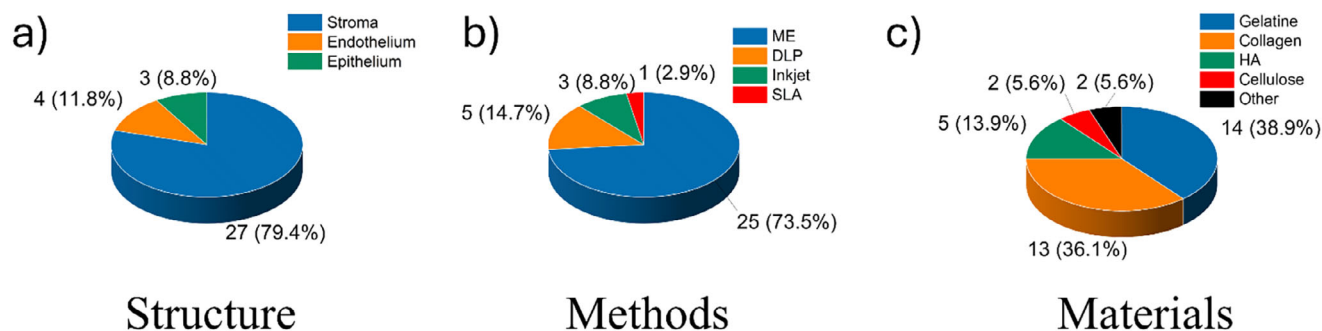


Figure 4. Classification of studies on 3D printed cornea by a) printed structure, b) printing methods, and c) printing materials. Percentages indicate the distribution of studies in each category. Abbreviations: ME, material extrusion; DLP, digital light processing; SLA, stereolithography; HA, hyaluronic acid. (Dual structures or printing methods were counted independently).

and 11 mm. In terms of thickness, it gradually increases from the centre to the periphery of the cornea. The average central thickness is 0.5–0.6 mm, and the periphery thickness is 0.6–0.8 mm, respectively.^[13] Among the whole 5 layers of the cornea, the stroma is the thickest layer, accounting for $\approx 90\%$ of the corneal thickness.^[14] It provides the strength to ensure cornea structure integrity and facilitates the transmission of light to the retina.

The stroma is also the most challenging layer to be 3D printed, and 79.4% of the 33 published papers have focused on stromal printing. It has a unique structure that provides transparency, the right curvature to refract the light and is mechanically strong. It is still not entirely clear how stroma retains all these properties; however, there is consensus that the ability to retain transparency and mechanical properties is closely linked to its well-organized collagen arrangement.

The stroma, it is primarily composed of keratocytes and multiple layers of tightly arranged collagen type I and V fibres, each at around 30 nm diameter (Figure 3b).^[15,16] The collagen fibres are arranged in layers, and each layer has a thickness ranging 1.5–2.5 μm .^[17] The stroma is made up of up to 200 of these layers.^[17] While the collagen fibres within each layer are parallel, their orientation alternates in orthogonal directions between adjacent layers (Figure 3b). Besides, the anterior third of the stroma exhibits a more isotropic fibre arrangement with higher fibre density, while the middle and posterior thirds have progressively more aligned fibres but lower density (Figure 3b,c).^[18,19] This structural gradient is attributed to greater exposure of the anterior layers to the external environment, creating a rigidity gradient that contributes to the cornea's dome shape.^[20,21] This precise arrangement of collagen fibres and their relatively low water content (75% dehydrated) was thought to be a key factor in corneal transparency. It was reported that this arrangement could minimise light scattering and enable incident light to interfere constructively, transmitting primarily in the forward direction.^[1,14,22–24] This essential fibre arrangement is difficult to achieve through printing.

3. Current Progress of the 3D Printed Cornea

The development of 3D printed corneas has been explored with the aim of addressing the shortage of donor tissues and treating various ocular diseases.^[26] Analyzing the 33 published 3D printing cornea papers showed that most studies (79.4%) were focused

on stroma regeneration, as the stroma constitutes the thickest and most structurally integral layer of the cornea. Among the different printing methods, Materials extrusion (ME) has been the most widely employed, accounting for 73.5%, followed by Digital light processing (DLP) at 14.7%, inkjet printing at 8.8%, and stereolithography (SLA) at 2.9%. Gelatine has been the most frequently utilized biomaterial, totalling 14, followed by collagen (13), hyaluronic acid (HA) (5), and cellulose (2) (Figure 4). The selection of materials and printing methods plays a key role in the fabrication of 3D-printed corneas, determining the physicochemical properties of 3D-printed products. These findings indicate that both the selection of biomaterials and the choice of printing methods are central in shaping the physicochemical properties of the printed constructs. Discussion on the advantages and disadvantages of different bioink types and printing strategies has been previously extensively reviewed.^[2,4,27] In our review, we focus on the less addressed aspects: mechanical strength and transparency of the printed corneal products. These features are critical for maintaining corneal function, yet are less discussed. In addition, the structure that enables curvature shape, mechanical strength, and transparency of the printed cornea also determines cellular behaviour and the long-term performance of the construct.

3.1. Curvature

Curvature is a critical structural factor, as it directly influences light refraction, intraocular pressure distribution, and overall mechanical stability. Its convex geometry provides the majority of the eye's refractive power, enabling proper focusing of light onto the retina. Beyond optical function, curvature also influences cellular behaviour through variations in surface tension and substrate strain. Studies have shown that regions with greater curvature can induce increased cell adhesion and promote an elongated cell morphology, likely due to changes in local mechanical cues, whereas flatter areas tend to support more rounded cell shapes (Figure 5a).^[28] Curved surfaces pose inherent challenges for layer-by-layer fabrication. In slicing the 3D model, smooth curves are approximated by many small flat facets, introducing a chordal error: each layer steps slightly over the intended curve.^[29] As layer thickness and slice angle increase, these deviations grow, where sloped surfaces exhibit visible faceting. Among

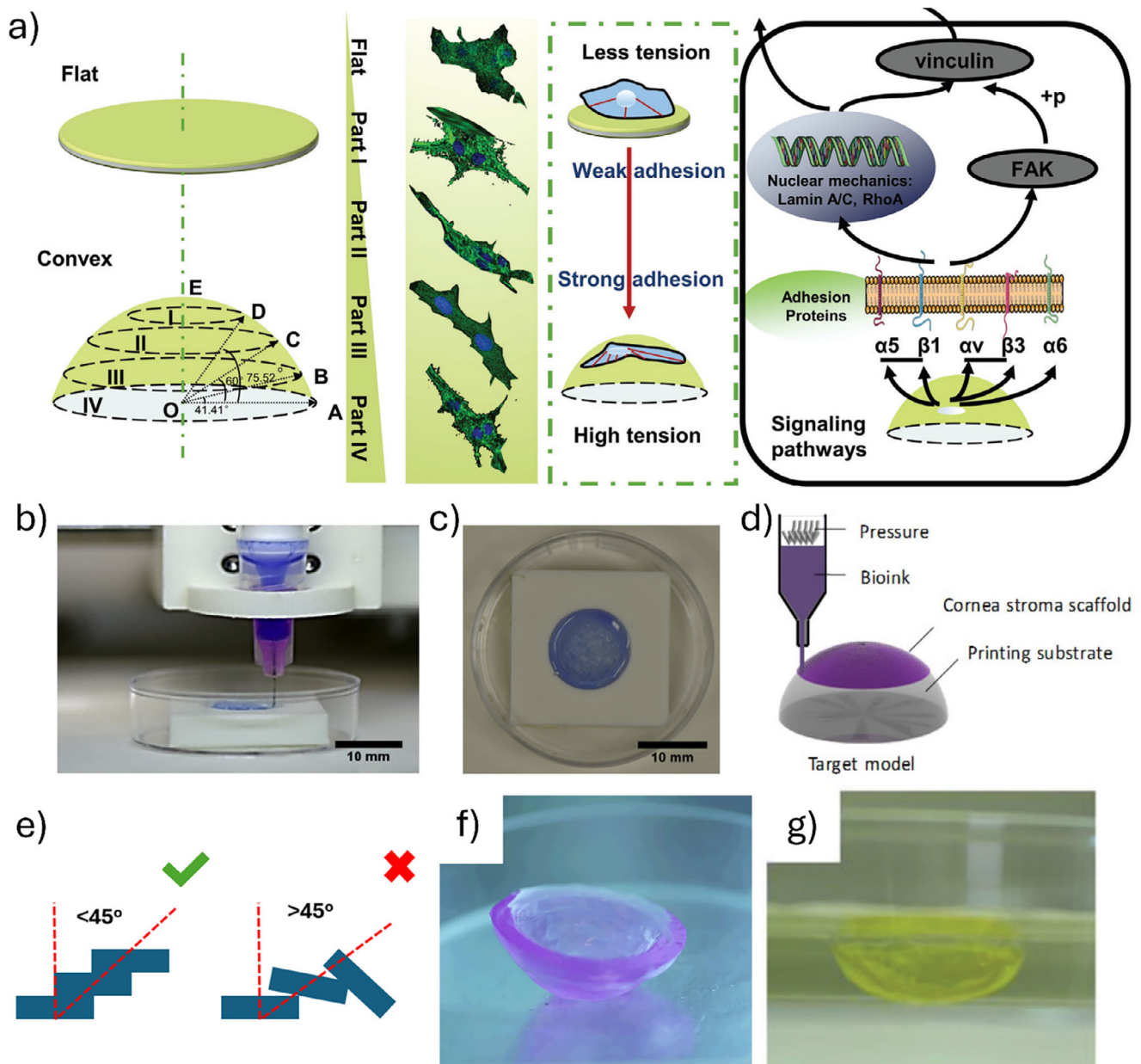


Figure 5. Strategies for printing stromal curvature and enhancing construct fidelity. a) Influence of slope angle on cell morphology and adhesion, with a schematic of underlying mechanotransduction pathways. **Reproduced with permission**^[28] Copyright 2023, Wiley-VCH GmbH. b) Side and c) top views of a concave support for stromal printing, and d) schematic of a convex support structure. **Reproduced with permission**^[30,31] Copyright 2018, Elsevier Ltd. and Copyright 2019, Zhejiang University and Springer-Verlag GmbH. e) Illustration of critical overhang angle for fabricating curved structures. f) Stromal curvature printed by ME without support structures. **Reproduced with permission**^[39] Copyright 2019, Wiley Periodicals, Inc. g) Reversed stromal construct printed by DLP. **Reproduced with permission**^[43] Copyright 2023, AccScience Publishing.

the 33 studies, only 13 them have been fabricated with a curved geometry, with most of these requiring support structures during printing to maintain their shape. Manufacturing curved stromal structures via 3D printing can be categorized into 2 approaches: mold-assisted fabrication and direct printing.

The mould-assisted approach is frequently combined with ME, as bioinks generally lack self-supporting ability, and gravitational forces limit the stability of freeform-printed structures. In this method, a concave or convex support structure is pre-

fabricated, onto which bioinks containing corneal stromal cells are extruded and subsequently crosslinked (Figure 5b–d).^[30–34] Once crosslinking is complete, the moulded stroma is detached for post-processing, including modification, sterilization, or drug loading.^[30,32,35] The mould-assisted method also allows for high-throughput printing, enabling the simultaneous fabrication of multiple corneas (6–12 per batch).^[36]

Direct printing of curved stromal structures is limited by the minimum printable overhang angle, which is typically limited

Table 1. Summary of the Young's modulus of the 3D printed cornea structure.

	Materials	Formulation of Ink	Methods	Structure	Young's Modulus	Reference
	Human Cornea				115.3 kPa	[48]
					270 – 520 kPa	[10]
					290 ± 60 kPa	[11]
1	Col-I	0.5% agarose-0.2% collagen	ME	Stroma	18.1 ± 3.5 kPa.	[39]
2	Col-I	6mg mL ⁻¹	ME	Stroma	47.18 kPa	[34]
		12mg mL ⁻¹			106.7 kPa	
3	Gelatine	1%alginate/10%gelatine	ME	Stroma	20.2 ± 0.4 kPa	[33]
4	Gelatine	10%GelMA/0.6%collagen	ME	Stroma	130 kPa	[28]
5	Gelatine	10%GelMA/1%CECM	DLP	Stroma	26.68 kPa	[43]
6	Gelatine	10%GelMA/0-4%eDCSM	DLP	Epithelium/stroma	19.56 ± 5.90–42.70 ± 3.93 kPa	[49]
7	Gelatine	15%GelMA	ME	Stroma	10 kPa	[37]
8	Gelatine	10%GelMA/1%sodium alginate	ME	Stroma	39.82kPa	[31]
		10%GelMA/2%sodium alginate			84.68kPa	
		10%GelMA/4%sodium alginate			178.43kPa	
9	Gelatine	5%GelMA/20%PEGDA(8k)	DLP	Epithelium/stroma	100.7kPa	[38]
		5%GelMA/15%PEGDA(8k)			35.9kPa	
		5%GelMA/10%PEGDA(8k)			13.9kPa	
10	Gelatine	5%GelMA	ME	Stroma	2.49kPa	[50]
		10%GelMA			30.01kPa	
		15%GelMA			60.36kPa	
11	Gelatine	20%Gelatin/2Alginat	ME	Epithelium/stroma	97.72kPa	[51]
		20%Gelatin/2Alginat/Ca2+			58.43kPa	
		20%Gelatin/2Alginat/Ca2+/EDC			82.99kPa	
12	Gelatine	5%GelMA/8%MC	ME	Stroma	71kPa	[52]
		7%GelMA/8%MC			118kPa	
13	Cellulose	10%CMCMA/5%HATA	ME	Stroma	190kPa	[35]
		10%CMCMA/20%HATA			237kPa	

to 45° (Figure 5e). This restriction ensures that each newly deposited layer has adequate support from the underlying layer to maintain structural stability. Initial attempts at direct stromal printing primarily involved printing planar^[7,37,38] or high-curvature structures^[28,39] to assess biocompatibility and post-printing cell behaviour (Figure 5f).^[40–42] The introduction of inverted DLP-based printing improved the fabrication of curved stromal constructs by counteracting gravitational effects, facilitating the successful printing of crescent-shaped artificial corneas (Figure 5g).^[43]

3.2. Mechanical Properties

A hydrogel with mechanical properties similar to native corneal tissue is essential for maintaining the shape and curvature of the cornea. Meanwhile, the cornea is continuously exposed to external forces, such as blinking, rubbing, and fluctuations in intraocular pressure, which challenge the long-term stability and integrity of the 3D printed cornea. 3D printed hydrogel with suitable mechanical strength not only enhances the biocompatibility of the hydrogel, but also reduces the risk of rejection or adverse reactions by providing a physiological environment for corneal cells.^[44] Hydrogel stiffness can guide cell behaviours, including

cell adhesion, migration, proliferation, and differentiation.^[45–47] Additionally, adequate mechanical robustness, particularly suturability, is necessary to withstand surgical handling without compromising the printed structure. The Young's modulus is a representative parameter to determine the resistance of a material to elastically deform under an external force.

We summarized the current Young's modulus of 3D printed cornea parts with materials, formulation, and printing methods (Table 1). The human cornea has a Young's modulus between 115 and 520 kPa. Various types of materials have been applied for 3D printing the cornea to mimic the mechanical strength of the human cornea. However, no reported 3D-printed cornea has achieved the upper end of this range. The highest recorded Young's modulus (237 kPa) was obtained using cellulose-based materials [10% carboxymethyl cellulose methacrylate (CMCMA) and 20% hyaluronic acid–tannic acid (HATA)].^[35] Other formulations that achieved values slightly above 100 kPa include printed high-concentration collagen (12 mg mL⁻¹),^[34] GelMA-collagen (10% GelMA/0.6% collagen),^[28] GelMA-sodium alginate (10% GelMA/4% sodium alginate),^[31] and GelMA-methylcellulose (7% GelMA/8% MC).^[42]

Amongst these materials, cellulose-based materials have much stronger strength due to the intermolecular hydrogen bonds that create a highly ordered and tightly bound crystalline

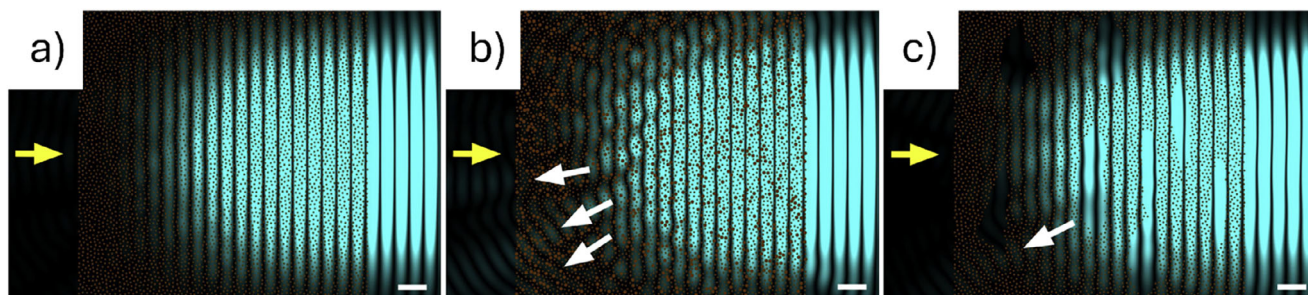


Figure 6. Secondary waves from a collagen fibril distribution presenting short-range order. a) Primary incoming light is travelling from left to right (yellow arrow), with a wavelength of 500 nm. Collagen fibrils in transverse sections are represented by brown circles. All fibrils have the same diameter of ≈ 31 nm, and no collagen fibrils can be closer than 62 nm. Only the intensity of the secondary radiation arising from the fibrils is shown in blue. No backward secondary radiation can be seen in the figure. b) The effect of increased fibril diameters on light transmission, and c) The effect of fibril voids on light transmission. Backward secondary radiation is evident in the figure (white arrows). Scale bar, 500 nm. Reproduced with permission^[1] Copyright 2015, Elsevier Ltd.

network. This structure enhances tensile strength but reduces hydrophilicity compared to collagen and gelatin, leading to lower moisture absorption. However, cellulose is not a native component of the cornea. In contrast, collagen is the primary structural protein in the native cornea, providing both mechanical strength and a supportive extracellular environment for corneal cells. Its original fibrillar arrangement in the cornea is essential for maintaining transparency and biomechanical integrity. GelMA, a derivative of gelatin, retains some structural similarities to collagen, including cell adhesion sites and biodegradability, making it a promising biomaterial for corneal printing. Nevertheless, these reported studies also confirmed that increased concentrations of biomaterials in the bioink will increase the strength of the printed structure. These findings suggest that the biomaterial used with current fabrication methods could have its inherent limit of mechanical strength. Even though a particular type I collagen has been used, it seemed to reach a limit at around 100 kPa. This mechanical strength gap between natural and 3D printed cornea might be highly likely due to the fibre alignment, which was hardly talked about or mentioned in the current study of 3D printed structure. Other methods are required to further increase the mechanical strength of printed structures.

3.3. Transparency

The functional cornea must maintain high optical transparency across the visible spectrum (400–700 nm) while also providing UV protection to prevent phototoxic damage to underlying ocular tissues. Achieving optimal transparency in 3D-printed corneal constructs requires precise control over collagen fibre arrangement, refractive index matching, and cellular composition.^[1] The stromal layer, which constitutes 90% of the corneal thickness, plays a dominant role in light transmission due to its highly organized collagen architecture.

Incident light interacting with collagen fibrils produces secondary spherical waves, which interfere constructively to form a forward-directed cylindrical wave (Figure 6a). Irregular fibril diameters or spacing increase backward radiation, causing light scattering and reducing transparency. (Figure 6b,c). Other fac-

tors influencing light propagation include the refractive index contrast between collagen and the surrounding extracellular matrix, the incident light angle, and secondary wave orientation.^[1,53] Maintaining precise control over these parameters is essential to preventing light scattering and ensuring high optical clarity in artificial corneas.

Transparency is commonly assessed by placing the printed construct over printed text or a grid and visually inspecting clarity.^[40] Quantitative transparency measurements, however, require spectrophotometric analysis to determine light transmittance at various wavelengths while accounting for absorption and scattering effects. Table 2 summarises the transparency values of 3D-printed corneas across different wavelength ranges, highlighting the impact of biomaterial formulation and printing techniques on optical properties. More studies report transparency data than mechanical strength data. Transparency values across different studies suggest that the use of Collagen-IV (12 mg mL^{-1}) to print structures demonstrates the highest optical clarity, exceeding 97% across the visible spectrum. Collagen-I (12 mg mL^{-1}), HA-DA (30%), DECM (2%), and HAMA (10%) also achieve over 90% transparency at longer wavelengths, indicating their strong potential as corneal substitutes.^[28,31,34]

In contrast, PVA (13%) exhibits significantly lower transparency ($\approx 62\%$ across all wavelengths), making it a less favourable choice for corneal applications. Gelatine-based formulations display varied transparency depending on their composition; for instance, 10% GelMA/2% sodium alginate reaches 97.3% at 700 nm, whereas 10% GelMA alone only reaches 88.68% at 700 nm. Nanocellulose-based constructs (80% nanocellulose/20% alginate) also show promising transparency ($\approx 98.91\%$ at 700 nm), suggesting their optical potential.^[52]

While collagen and gelatin-based hydrogels exhibit superior optical performance, they often require additional modifications to enhance mechanical properties. The findings highlight that while increasing biomaterial concentrations in inks can improve mechanical strength, achieving both optimal transparency and biomechanical performance remains a challenge. Further optimization of collagen fibril alignment and refractive index matching may enhance the optical properties of 3D-printed corneas.

Table 2. Summary of the transparency of the 3D printed cornea at a specific wavelength.

Materials	Formulation of Ink	Methods	Structure	Transparency [%]							Reference
				300	400	450	495	570	620	700	
Human cornea				N/A	65.25	70.99	74.1	77.21	78.88	80.31	[38]
				52	85.4	89.8	92.2	95.6	95.4	97.6	[40,54]
				N/A	76.8	83.9	87.26	93.09	94.3	94.1	[43]
1	DECM	2% DECM		N/A	82.03	86.52	88.6	91.31	92.35	93.7	[1]
2	Gelatin	10% GelMA/ 2% sodium alginate	ME	75.7	97.3	97.6	98	97	97.6	98.4	[40,54]
3	Gelatin/ collagen	4% Gelatin/ 3.25% alginate/ 5mg mL ⁻¹ collagen	ME	47.88	84.42	90.76	93.65	95.19	96.53	97.3	[31]
			ME	61.83	77.09	79.77	82.44	85.49	88.93	90.83	[36]
4	PVA	13% PVA	ME	N/A	43.82	61.89	62.2	62.31	62.52	62.73	[55]
5	Gelatin	5% GelMA/ 20% PEGDA(8k)	DLP	N/A	71.23	76.01	78.16	81.03	82.23	84.86	[38]
6	HA-DA	30% HA-DA	ME	N/A	73	81.74	85.74	90.58	94.1	94.48	[56]
7	Col-I	12mg mL ⁻¹	ME	N/A	92.85	95	95.75	95.88	96.04	95.16	[34]
8	Nano cellulose	80% nanocellulose/ 20% alginate	ME	38.1	79.68	86.23	89.93	94.96	97.14	98.91	[57]
9	Gelatin	10% GelMA	ME	2.74	65.23	80.76	84.87	86.7	87.9	88.68	[50]
10	CECM- GelMA	10% GelMA/ 1% CECM	DLP	N/A	54.6	62.5	74.7	80.36	81.37	80.16	[43]
11	HAMA	10% HAMA/ 1% lithium benzoyl (phenyl) phosphinate/ tartrazine	DLP	N/A	90.63	93.23	94.34	95.67	96.23	96.81	[58]
12	Gelatin	20% Gelatin/ 2Alginat/ Ca2+/ EDC	ME	N/A	75.39	78.79	81.14	83.5	84.61	87.63	[51]
13	Col-IV	12mg mL ⁻¹	ME	N/A	96.96	97.4	97.68	98.19	98.15	98.07	[59]
14	HA	9mg mL ⁻¹ HA-DA-CDH/ 9mg mL ⁻¹ HA-ALD	ME	N/A	76.26	84.67	90.4	95.45	96.96	96.96	[44]
15	Gelatin	5% GelMA/ 8% MC	ME	0	58.43	66.1	70.02	72.45	72.71	74.07	[52]
16	Gelatin	10% GelMA/ 0.5% agarose	ME	0	60.34	71.42	71.92	72.9	73.39	73.53	[60]

Table 3. Key characteristics of the human cornea with the standard measurement method compared with 3D printed cornea constructs.

Parameter	Standard measurement method	Typical value (Human Cornea)	Current published study	Reference
Young's Modulus	- Atomic force microscopy - Uniaxial/biaxial tensile testing - Strip extensometry	115–520 kPa	Refer to Table 1	[10,48]
Transparency	UV–vis spectrophotometry	>90% (visible range, 400–700 nm)	Refer to Table 2	[1,40,54]
Refractive Power	- Keratometry - Optical coherence tomography	≈43 dioptres (total corneal power)	No published result	[69]
Refractive Index	- Optical coherence tomography - Abbe refractometry	≈1.376 (total corneal index)	GelMA 1.387 Type I and type III collagen 1.345 to 1.355	[32] [4]
Corneal Permeability	- Fluorophotometry - Diffusion chambers (e.g. Franz cell) - Using chamber	≈10^{−10} (cm² s^{−1}) [mL O₂·cm^{−3} mmHg^{−1}] (for oxygen) ≈10^{−6}–10^{−7} cm s^{−1} (for small hydrophilic molecules)	HA-dopamine 1.343 No published result	[56] [62,66,67]

3.4. Key Corneal Properties not Adequately Reported

Although most studies on 3D-printed corneas have evaluated mechanical strength and transparency, several other biomechanical properties are underreported. Refractive power indicates the focusing ability of the cornea, which is determined by the materials used and the overall curvature of the construct. The refractive index of the material directly affects light refraction; thus, matching the refractive index of 3D-printed corneas to that of native tissue is crucial for minimizing glare, light scattering, and optical aberrations. Common methods for measuring the refractive index include the critical angle method, immersion techniques, hollow acute prism, and Abbé refractometry. The reported refractive index of 3D printed materials is GelMA 1.387,^[32] type I and type III collagen 1.3451–1.3552,^[4] and HA-dopamine 1.343.^[56] The overall average refractive index of the human cornea is 1.376, contributing to the corneal refractive power of 43 dioptres.^[8] However, few studies provide comprehensive data on the refractive index or refractive power of printed constructs.

Another parameter that is frequently overlooked is corneal permeability. The native cornea relies on passive diffusion of oxygen from the tear film and aqueous humour, as well as glucose transport, to sustain metabolic activity. On average, the oxygen tension in the aqueous humour is ≈24.1 mmHg, while at the interface between the cornea and the atmospheric air, the value is ≈155 mmHg under open-eye conditions, and ≈61.5 mmHg from the palpebral conjunctiva under closed-eye conditions. The minimum precorneal oxygen concentration required for aerobic respiration and preventing corneal swelling is 10% (approximately oxygen tension of 74 mmHg).^[61–63] Lower oxygen levels can cause hypoesthesia, acidosis, epithelial microcysts, stromal and epithelial thinning, increased endothelial polymegathism, hypoxia-induced keratocyte activation, stromal fibrosis, and loss of corneal transparency, while excessive permeability may be due to a compromise in the mechanical and optical integrity of the construct.^[64,65] Therefore, the materials selected for biofabricated corneas must possess sufficient permeability to facilitate the diffusion of oxygen (49.2×10^{-11} (cm²s^{−1}) [mL O₂·cm^{−3} mmHg^{−1}]), glucose ($3.0 \pm 0.2 \times 10^{-6}$ cm²s^{−1}), and other essential

nutrients.^[62,66,67] This aspect remains largely unexplored in current studies and represents a critical area for future research.^[67,68]

3.5. Fabrication Challenges using 3D Printing

Despite numerous studies reporting on key characteristics such as mechanical strength and transparency, the standardization of the properties of 3D-printed corneal constructs remains inconsistent. Many published papers provide data on Young's modulus via tensile or compression testing, and light transmittance via UV–vis spectrophotometer; however, several critical parameters—including refractive power, refractive index, and corneal permeability—are rarely reported (Table 3). For instance, while refractive index measurements are essential in defining the focusing ability of the cornea and ensuring optical fidelity, only a few studies have reported these values of the materials they applied. Similarly, the permeability of the printed constructs to oxygen and glucose, which is vital for maintaining cell viability and metabolic activity, has not been consistently addressed. This lack of standardized characterization hinders direct comparisons between studies and the advancement of clinically viable constructs.

The previous sections highlight key challenges of the difficulty in achieving both high mechanical strength and optimal transparency. One contributing factor is that the native human cornea is composed of highly ordered, orthogonally aligned fibrous collagen architecture. This organized collagen network is critical for both optical clarity and biomechanical performance.^[1] In contrast, current 3D printing techniques have not achieved to replicate this precise fibre alignment. For example, conventional extrusion-based printing typically results in randomly distributed collagen fibres, which increase light scattering and diminish transparency.^[54] Moreover, strategies aimed solely at increasing the concentration of primary bioink components or incorporating reinforcing agents such as PEGDA, alginate, or cellulose have only yielded modest improvements in the mechanical strength of bioengineering cornea.^[31,34,52] These approaches typically lack organized fibre architecture, which limits their ability to replicate the mechanical resilience of native corneal tissue

and diminishes their capacity to withstand physiological forces such as intraocular pressure and external mechanical stress. To enhance the mechanical strength, some studies have introduced oriented microstructures into hydrogels. These strategies enhance anisotropic mechanical properties by aligning fibres to serve as load-bearing elements within the hydrogel matrix, significantly improving tensile strength along the alignment direction, yet a gap remains between engineered and natural corneal tissue.^[70–72]

Another major challenge in 3D-printed corneal constructs is the fabrication of a naturally curved stromal structure without relying on external support materials. The human cornea exhibits a dome-like curvature essential for light refraction and visual acuity. However, most current 3D printing techniques rely on flat or scaffold-supported printing, which can lead to structural inconsistencies when attempting to fabricate curved constructs. The use of support materials often complicates the printing process, requiring additional post-processing steps for removal, which may compromise the overall structural integrity and interlayer adhesion. Alternative printing strategies that enable freeform, support-free curved fabrication are essential to overcoming these limitations. Several advanced 3D printing technologies, discussed below, may offer an approach to address these challenges.

4. Advanced Technologies in 3D Printing

Over the past three years, breakthroughs in 3D printing have dramatically advanced the field of tissue engineering. These innovations have led to the development of novel techniques such as ultrasound-assisted printing, magnetic field-assisted printing, volumetric additive manufacturing and continuous liquid interface production (Figure 7 and Table 4). These recent advancements are expected to play a crucial role in overcoming longstanding challenges in replicating the native cornea's structural fidelity and ultimately paving the way for the required functionality.

4.1. Advanced Strategies for Precise Fibre Alignment in 3D-Printed Corneas

Replicating the native corneal stroma requires precise control over collagen fibre deposition and alignment. Techniques such as ultrasound-assisted collagen orientation and magnetic field-assisted printing hold significant promise in this regard.

Ultrasound-based printing employs focused ultrasound waves to form the aimed pattern, triggering sonochemical polymerization (Figure 7a,b).^[73–75] This approach enables the use of heat-curing thermosets, which is challenging in traditional photo- or thermally induced polymerization. Thereby, ultrasound-assisted printing expands the range of biocompatible biomaterials, holding potential for fabricating corneal grafts with integrated biosensing functionalities.^[76] Meanwhile, it also overcomes limitations associated with in situ deep layer printing, especially when bioinks contain highly scattering particles, by facilitating non-invasive, remotely controlled deposition.^[75] In addition, acoustic waves can modulate the air–liquid interface dur-

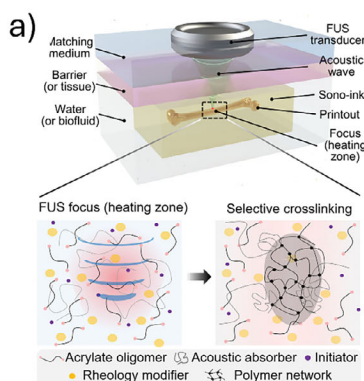
ing printing, thereby enhancing structural resolution and fabrication speed (Figure 7c).^[77] By dynamically adjusting the pressure within the printhead, the curvature and position of air–liquid printing boundary can be precisely controlled, improving the printing resolution and speed, and minimizing the scattering and light absorption from the suspended cells or other particles.^[77] Moreover, the acoustic standing waves at the liquid–gas interface can generate periodic pressure fields that induce the alignment and aggregation of suspended particles or cells at defined nodal positions. When applied to bioinks containing collagen molecules or fibrils, these pressure nodes can guide their orientation along specific directions, promoting anisotropic fibre alignment. This alignment mimics the organized collagen architecture of the native corneal stroma, thereby improving the mechanical and optical properties of the printed construct, both of which are crucial for engineering functional corneal tissues.^[78]

Magnetic force is another external stimulus used to assist in guiding cells or biomaterials in a specific direction. When magnetic nanoparticles, such as iron oxide-based, are mixed into the bioink, they enable the formation of anisotropic, aligned structures under the influence of an external magnetic field (Figure 7d).^[79–81] These nanoparticles respond to the magnetic field by orienting along its lines of force, thereby physically directing the distribution of surrounding matrix components. In composite hydrogels containing collagen, the magnetic alignment of embedded nanoparticles can influence the orientation of collagen fibrils by mechanical tethering or through magnetic manipulation of nanoparticle–collagen interactions. This process results in aligned collagen microstructures, which better replicate the lamellar architecture of the native corneal stroma. Moreover, the applied magnetic field induces remote, non-contact magneto-mechanical stimulation, which can affect the extracellular matrix and cytoskeletal organization of encapsulated cells. This mechanical input promotes cell alignment and activates mechanotransduction pathways involved in corneal tissue development.^[82–85]

4.2. Strategies for Support-Free Curved Corneal Constructs

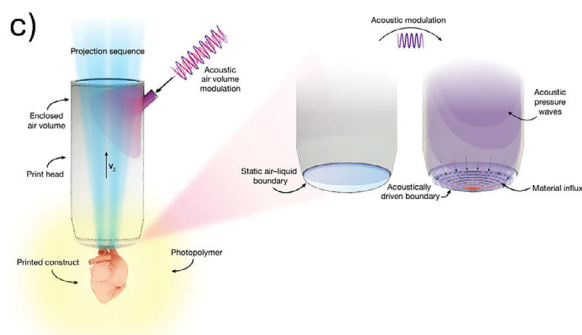
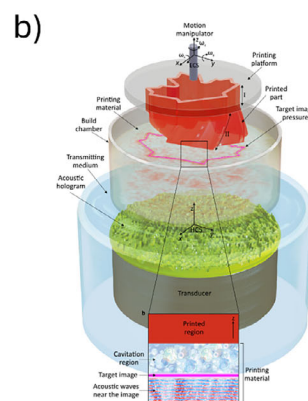
In addition to fibre alignment, freeform printing strategies that enable support-free curved fabrication are essential for creating anatomically accurate corneal constructs. Techniques such as ultrasound-assisted printing, volumetric additive manufacturing (VAM), and xolography are emerging as promising solutions (Figure 7e,f).

VAM was first reported by Shusteff et al. in 2017, using the superposition of two or more light beams to solidify the resin.^[86] Later, Kelly et al. refined the system to tomographic VAM, which is now widely adopted due to its ability to fabricate overhanging and spanning geometries without the need for support structures.^[87] By eliminating the need for external supports, VAM addresses issues related to poor surface quality and directional differences in mechanical properties. Besides, it enables the printing of external components onto the pre-existing structure with minimum mechanical stress on the printed part.^[88] One of the most promising applications of VAM in corneal printing is its potential to eliminate interface-related inconsistencies between adjacent corneal layers. By enabling simultaneous volumetric fabrication of multiple tissue layers, VAM



Acoustic volumetric printing applies sound waves creating standing or traveling acoustic fields within a liquid medium. The sonothermal effect triggers the decomposition of the initiator for local polymerization.

Holographic direct sound printing combines holography and acoustic waves to create and manipulate structures in three dimensions within a medium. The method uses the interference of ultrasound waves to control the positioning of particles or droplets in space, similar to how holography controls light to create 3D images.



Dynamic interface printing is based on an air–liquid boundary forms at the base of a partially submerged print head. The boundary acts as a print interface in which patterned projections are used to locally solidify the photopolymer.

Magnetic field-assisted bioprinting use external magnetic fields to guide the assembly and patterning of bioprinted materials. This process typically involves incorporating magnetic particles into bioinks. The cells or biomaterials can be manipulated by the magnetic field during the printing process to achieve precise control over the structure and alignment of the printed tissue or scaffold

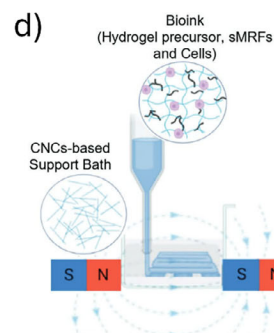
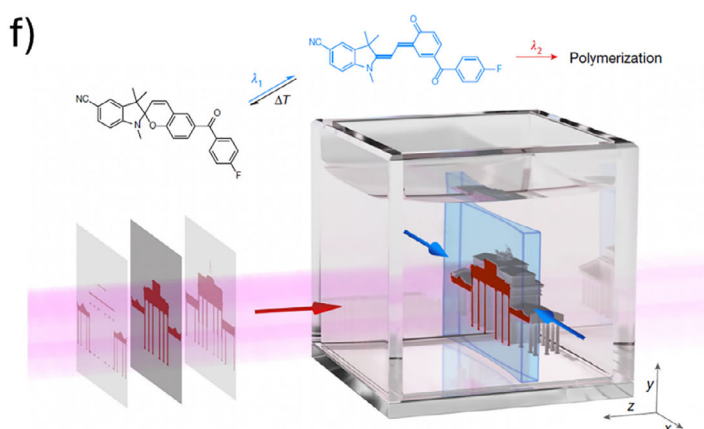
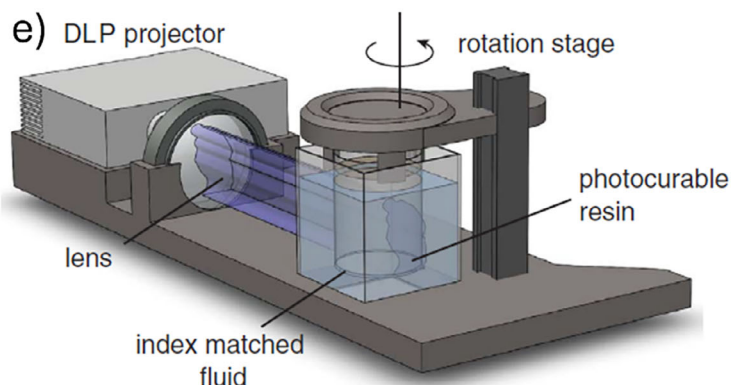


Figure 7. Representative advanced 3D printing strategies to overcome current limitations in corneal bioprinting: a) Acoustic volumetric printing. Reproduced with permission^[75] Copyright 2023, The American Association for the Advancement of Science. b) Holographic direct sound printing. Reproduced with permission^[74] Copyright 2024, Springer Nature. c) Dynamic interface printing. Reproduced with permission^[77] Copyright 2024, Springer Nature. d) Magnetic field assisted printing. Reproduced with permission^[80] Copyright 2022, Wiley-VCH GmbH. e) Volumetric additive manufacturing. Reproduced with permission^[87] Copyright 2019, The American Association for the Advancement of Science. f) Xolography. Reproduced with permission^[90] Copyright 2020, Springer Nature, and g) Injection continues liquid interface production. Reproduced with permission^[12] Copyright 2022, The American Association for the Advancement of Science.

Volumetric additive manufacturing employs light energy to deliver a series of 2D images to a rotating resin vat. In the printing process, Radon transform was used to convert 3D model into multiple 2D images. These 2D images could represent the 3D model from different perspectives. An inverse Radon transform calculates the 3D distribution of light intensity from projectors, completing the printing process.



Injection continues liquid interface production (iCLIP) continuously injected resin into vat during printing process. Meanwhile, a continuous dynamic liquid interface is maintained between resin and solid substrate, allowing oxygen supply to form smooth layer.

Xolography applies two orthogonally beams of light with different wavelength focusing on resin volume. Only crossed light projected area initialize the polymerization process. With the synchronized movement of the resin volume, a sequence of images projected and form the aimed object.

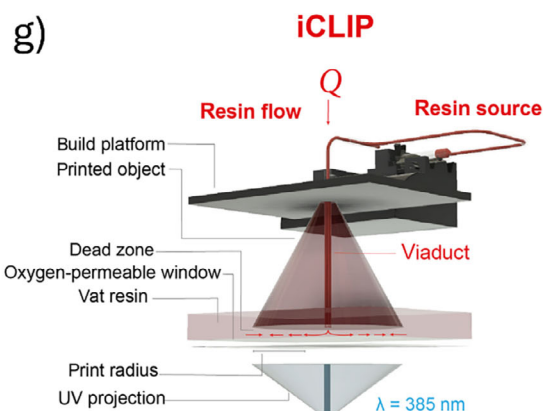


Figure 7. Continued

addresses the assembly challenges associated with multi-layer corneal constructs, ensuring both mechanical stability and optical clarity.^[31,89–92] VAM is also applicable to collagen-based fabrication, which is similar like gelatine based metamaterial has high viscosity under low temperature and is crosslinkable by mixing photoinitiator under a specific wavelength of light.^[34,59] This adaptation would allow the use of VAM to produce collagen-rich constructs with controlled microarchitecture and minimal interfacial defects. Additionally, because VAM operates without physical extrusion, it avoids shear-induced damage, which is espe-

cially beneficial for preserving the native structure of collagen fibrils.

Despite its advantages, VAM faces challenges in precise multi-material deposition within a single fabrication chamber.^[93,94] Multimaterial printing is essential not only to facilitate the co-deposition of various biomaterials, but also enables multimodal sensing capabilities, including tensile, compression, shear, torsion and light stimulation. Given the complex mechanical and optical stimuli response of the human cornea, multi-material printing strategies will be essential for fabricating functional

Table 4. Summary of the working principles, key features, and limitations of the representative advanced 3D printing methods.

Technology	Working principle	Key features	Limitations	Reference
Acoustic volumetric printing	Uses focused acoustic waves to induce localized polymerization volumetrically	• Remote, non-invasive patterning • Allows printing in deep tissue or scattering media • Suitable for soft materials	• Limited material compatibility • Complex equipment • Resolution is limited by the ultrasound wavelength	[75]
Holographic direct sound printing	Combines holography with sound-induced polymerization for 3D patterning	• Enables patterned, deep 3D printing • Support free • High-speed volumetric print	• Complex hardware setup • Limited by acoustic-material interactions	[74]
Dynamic interface printing	Controls air–liquid interface with acoustic waves, in which a patterned projection is applied in photopolymer	• Enhanced resolution and speed • Minimized light scattering from particles	• Requires precise control systems • Not widely validated for thick tissues	[77]
Magnetic field-assisted printing	Guide the assembly and patterning of materials under external magnetic fields	• Induces anisotropic structures and cell alignment • Allows remote control	• Alignment depends on particle-matrix interaction • Biocompatibility concerns	[80]
VAM	Uses tomographic light projection to solidify entire volumes without layers	• No support structures needed • High geometric complexity • Layer-free	• Limited multi-material capability • High equipment cost	[87]
Xolography	Uses intersecting light sheets in a photo-reactive medium for 3D fabrication	• High resolution • Fast fabrication • Reduced interface artifacts	• Material constraints • Scalability issues	[90]
iCLIP	Uses microfluidics to feed multiple bioinks into a photopolymerization zone	• Multi-material continuous printing • High throughput • Good interface integration	• UV compatibility required • Microfluidic integration complexity	[12]

bioengineered corneas.^[95] Injection continuous liquid interface production (iCLIP) can potentially meet the multi-material requirements, utilizing microfluidic channels to continuously feed bioinks into the printing interface. This approach supports high throughput, continuous multimaterial printing and holds great potential for fabricating full-thickness, multi-layered corneal constructs in a single fabrication step (Figure 7g).^[12]

By integrating these advanced 3D printing strategies, future research would potentially overcome existing limitations in corneal biofabrication. The development of clinically viable, full-thickness artificial corneas that closely mimic the structural, biomechanical, and optical properties of native tissue represents the next frontier in ocular tissue engineering. These technological innovations are crucial in bridging the gap between current laboratory research and real-world clinical applications, ultimately leading to scalable solutions for corneal tissue engineering and transplantation. Collectively, these advancements provide a strong foundation for the future development of fully functional, patient-specific corneal grafts.

5. Outlook and Opportunities

The global shortage of corneal donors limits the number of corneal transplant surgeries, leading to prolonged waiting times and, in some cases, irreversible vision loss. Addressing this challenge requires the development of bioengineered corneal substitutes that can replicate the structural, mechanical, and optical properties of native tissue. 3D printing, as an emerging technology, offers precise control over tissue dimensionality, organization, and cell–cell and cell–matrix interactions, making it a highly promising approach for corneal tissue engineering. This capability is particularly relevant for fabricating anisotropic and structurally complex multi-tissue constructs.

A critical aspect in the development of 3D-printed corneal constructs is the enhancement of mechanical performance, as native corneal tissue must simultaneously provide optical clarity and sufficient biomechanical stability. Recent advances in bioink formulation and printing techniques have demonstrated significant potential, and the functionality and behaviour of biomaterials encapsulated cells have been widely observed and investigated. With the challenges associated with cell encapsulation largely addressed, the next critical step is to replicate the mechanical properties of the native cornea. Establishing an appropriate biomechanical environment is essential to guide cellular behaviour and ensure the long-term functional performance of 3D-printed corneal constructs. Although significant progress has been made in reproducing macroscopic structures, the hierarchical organization of collagen fibrils still requires further optimization. Achieving an interwoven, orthogonal alignment of collagen fibres is necessary to replicate both the biomechanical and optical properties of native tissue.^[96–98] Conventional layer-by-layer fabrication might have limited ability to arrange printed fibres. Emerging technologies, such as volumetric printing, iCLIP, and ultrasound-induced cell and collagen alignment, may assist in a rapid, hybrid materials applied and pattern-controlled way to provide a promising approach to mimic native stromal architecture.^[99–102]

To further address the inherent weakness of hydrogel-based bioinks, researchers have explored several strategies. One

promising approach involves the introduction of multiple physical and chemical crosslinking mechanisms within hydrogel systems. For example, dual crosslinking by combining ionic and covalent bonds can provide improved stiffness and durability while maintaining flexibility.^[103,104] Supramolecular interactions, such as host–guest assembly or hydrogen bonding, have also been utilized to generate reversible physical networks that enhance toughness and self-recovery.^[105] In parallel, enzymatic crosslinking methods offer biocompatible and tenable control over gelation, further contributing to structural reinforcement.^[106] In addition to in situ crosslinking strategies, post-printing reinforcement treatments have been proposed to strengthen the printed constructs without compromising their shape fidelity.^[107] Photocrosslinking using UV or visible light is widely applied to tighten polymer networks after deposition, resulting in improved load-bearing capacity.^[107] Chemical treatments with crosslinking agents, such as genipin or carbodiimide-based compounds, have been shown to further stiffen collagen- or gelatine-containing hydrogels.^[108,109] Moreover, physical reinforcement methods, such as dehydration or freeze–thaw cycling, can compact polymer chains and increase mechanical integrity.^[110–112]

While these reinforcement methods can significantly improve mechanical performance, some are associated with potential drawbacks, including cytotoxic effects from residual crosslinkers or damage to encapsulated cells during harsh treatments.^[113] For corneal applications where cell viability and transparency are critical, strategies that rely on intrinsic structural design or biological remodelling may be more suitable. Cell-mediated remodelling, in which encapsulated keratocytes or stem cells deposit extracellular matrix over an extended culture, has been proposed as a means of gradually increasing tissue strength while preserving cell viability and optical clarity.^[114] Similar approaches have been successful in other soft tissue engineering applications, suggesting a promising pathway for future corneal biofabrication.^[115,116]

Besides, printing a hollow curved dome structure for corneal regeneration is still challenging because curvature amplifies the intrinsic limitations of layer-wise and projection-based fabrication.^[49,52] In layer-by-layer extrusion (ME), the corneal surface is approximated by discrete deposition paths, producing a chordal or “stair-step” error that increases with layer thickness and local slope.^[29] Steep slopes and overhangs, therefore, show larger deviations from the design radius and are susceptible to sagging or collapse unless supported or moulded. Material factors that interact with curvature include bioink rheology (viscosity, shear thinning), gelation/sol–gel kinetics and surface tension; slow gelation or low yield stress increases risk of deformation during and immediately after deposition, while excessively high viscosity can impair filament continuity and resolution.^[52] In light-based printing, like DLP printing, the spreading of projected light could cause over-curing, which results in thicker walls and loss of fine curvature detail.^[117] The absence of internal support makes structural stability difficult to preserve, and resin entrapment or water evaporation warping can alter wall thickness.^[118] Maintaining dimensional accuracy in such designs remains a significant technical barrier, and careful control of layer thickness, build orientation, support strategies, and exposure settings is required to approach the intended geometry.

Despite these known issues, published studies rarely report quantitative geometry metrics.^[49,60] To our knowledge, no study

has systematically reported measured geometric errors (e.g. deviations in radius or thickness) as a function of curvature in such constructs. Including these data would greatly strengthen the analysis. For example, printing a series of test domes with controlled printing parameters and measuring their actual profiles via high-resolution scanning or optical coherence tomography would allow correlation of curvature angle to dimensional error. Key metrics should include radius error, local wall-thickness variation, surface roughness, and topographic chordal error. Incorporating empirical measurements of the printed geometry would provide concrete guidance for future corneal scaffold fabrication and design optimization. Furthermore, proposing a simulation model would be great to minimise efforts, adjust printing parameters and achieve possible high-precision DLP printing of photocurable materials.^[119]

In addition to mechanical strength and structural fidelity, other key properties, such as the refractive power and oxygen/nutrition permeability, are also key biomechanical properties for the cornea and have not been addressed. We advocate for a standardized reporting system that includes not only biocompatibility tests but also quantitative assessments of mechanical strength, transparency, refractive power, and oxygen permeability. Establishing such standards will enable effective comparisons across studies and accelerate the development of clinically viable bioengineered corneas.

In conclusion, while 3D printing has made significant strides in corneal tissue engineering, further research is needed to refine fibre organization, enhance mechanical robustness, and comprehensively evaluate all critical biomechanical properties. The integration of multi-modal crosslinking mechanisms with post-printing reinforcement strategies represents a promising pathway toward replicating the mechanical requirements of the human cornea. Future work should focus on balancing the trade-off between mechanical enhancement and optical transparency, as excessive crosslinking or nanomaterial loading may compromise light transmission. By addressing these challenges, the field can move closer to realizing patient-specific, full-thickness corneal grafts that offer a scalable, rapid, and cost-effective solution for restoring vision.

To bridge this gap, we should propose and develop a comprehensive, standardized characterization protocol for 3D-printed corneas. Such a protocol should include, at a minimum, assessments of mechanical properties (e.g., Young's modulus), optical transparency, refractive index, refractive power, and permeability. Establishing uniform evaluation criteria will facilitate more meaningful comparisons across studies and accelerate the translation of 3D printed corneal constructs into clinical applications.

Acknowledgements

The authors wish to acknowledge funding from the Medical Research Future Fund (RFRHP1000013), Australian Research Council Future Fellowship (FT230100249), Sydney Horizon Fellowship, and Biomanufacturing Incubator EMCR Seed Funding.

Open access publishing facilitated by The University of Sydney, as part of the Wiley - The University of Sydney agreement via the Council of Australian University Librarians.

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Y.Y. did writing of review & editing, writing of original draft, visualization, validation, software, and conceptualization. K.L. did the writing of the review & editing and conceptualization. G.S. did the writing of the review & editing and writing of the original draft. G.G.W. did the writing of the review & editing, writing of the original draft, and investigation. J.Y. did the writing of the review & editing, writing of the original draft, supervision, and conceptualization.

Keywords

3D printing, cornea, mechanobiology, regenerative medicine, transparency

Received: June 2, 2025

Revised: September 17, 2025

Published online: October 3, 2025

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